

Bitwise linear trail through **WARP** found by SAT

CiVerLy

July 21, 2026

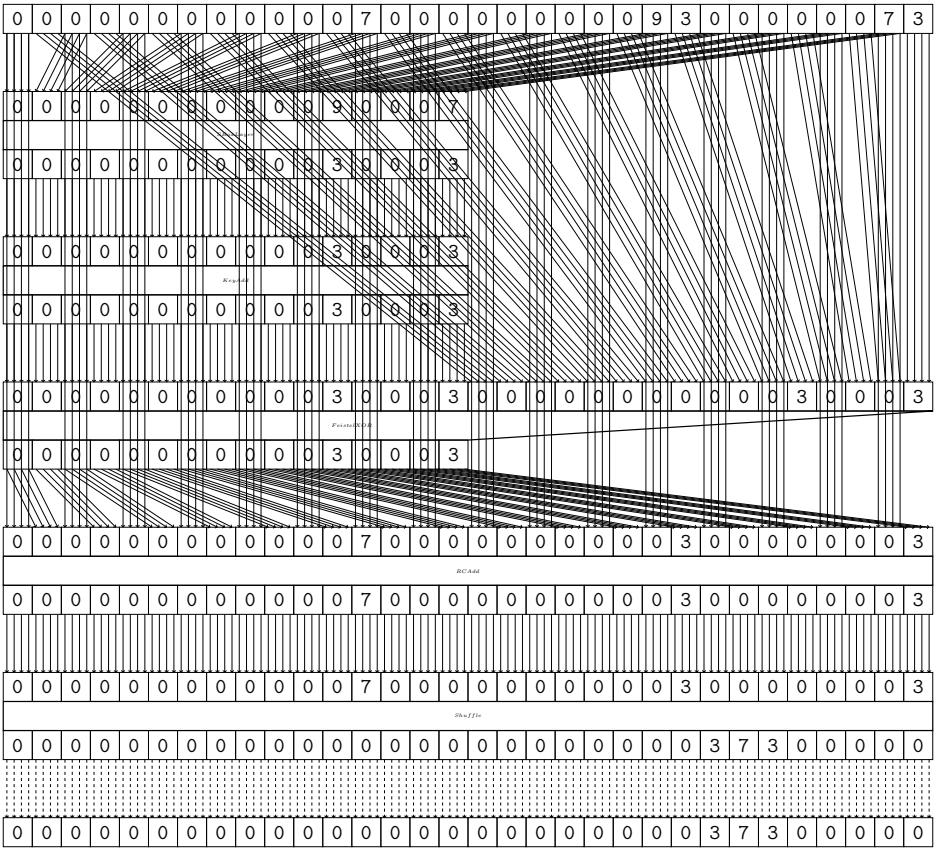
1 WARP

Maximal linear correlation: 2^{-11}

0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	9	3	0	0	0	0	0	7	3
0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	9	3	0	0	0	0	0	0	7	3
unpermut																														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	7	3	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	7	3	0	0	0	0	0	0	
unpermut																														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0	0	0	0	0	0	0	0	0	0	
unpermut																														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	
unpermut																														
0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0
unpermut																														
0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	
unpermut																														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	3	7	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	3	7	0	0	0	0	
unpermut																														
0	0	0	0	0	0	0	0	7	0	0	0	0	0	9	3	0	0	7	0	0	3	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	7	0	0	0	0	0	9	3	0	0	7	0	0	3	0	0	0	0	0	0	0	0	
unpermut																														
0	0	7	0	0	3	0	0	0	9	0	0	3	0	3	0	0	0	0	0	0	3	0	0	0	3	0	0	7	0	
unpermut																														
0	0	7	0	0	3	0	0	0	9	0	0	3	0	3	0	0	0	0	0	0	3	0	0	0	3	0	0	7	0	

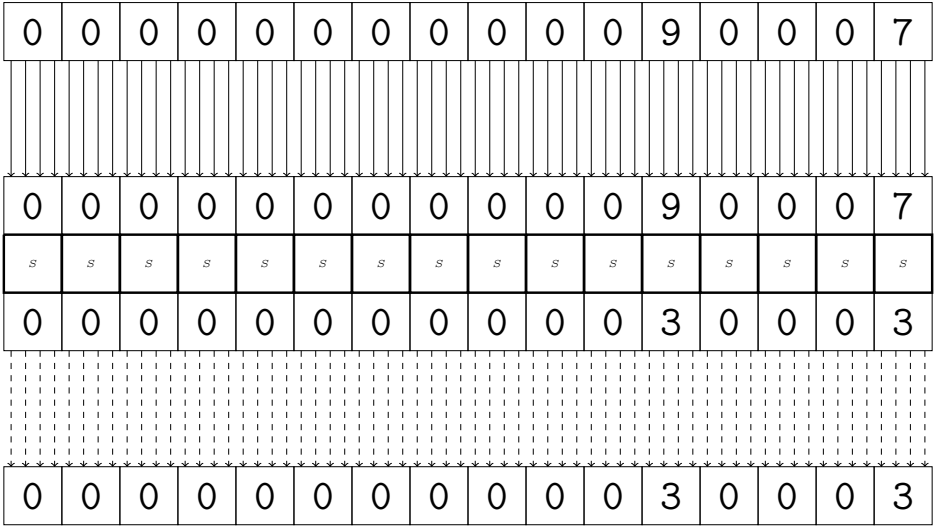
2 warp_round

Maximal linear correlation: $2^{-2.0}$



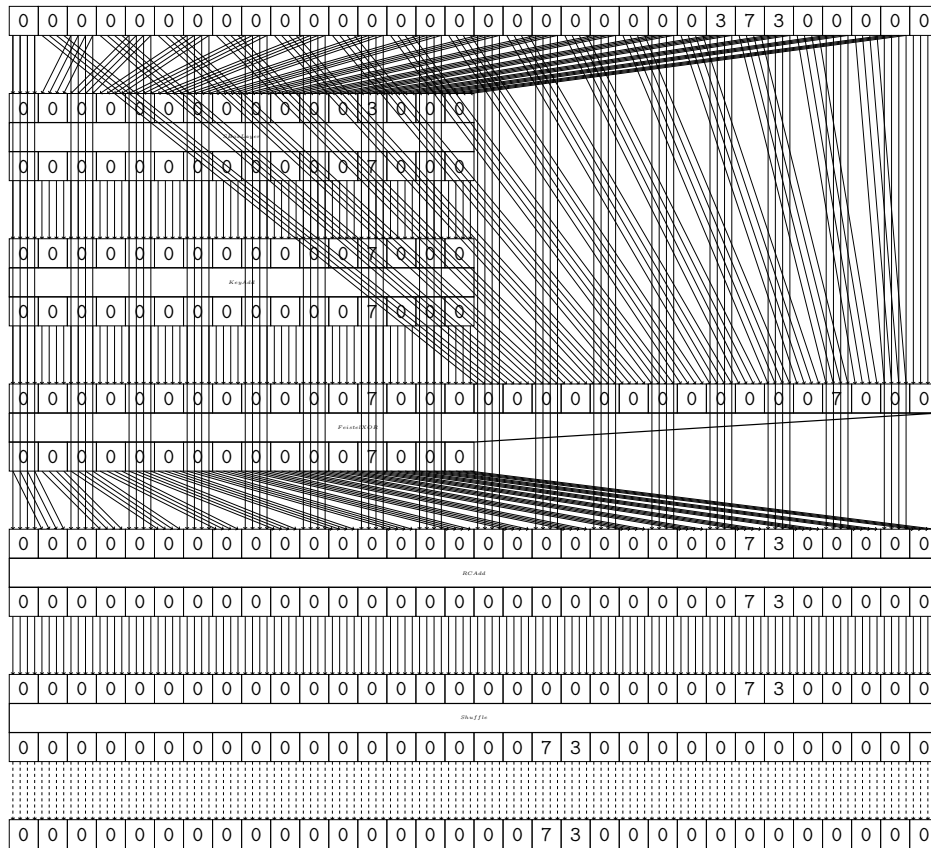
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



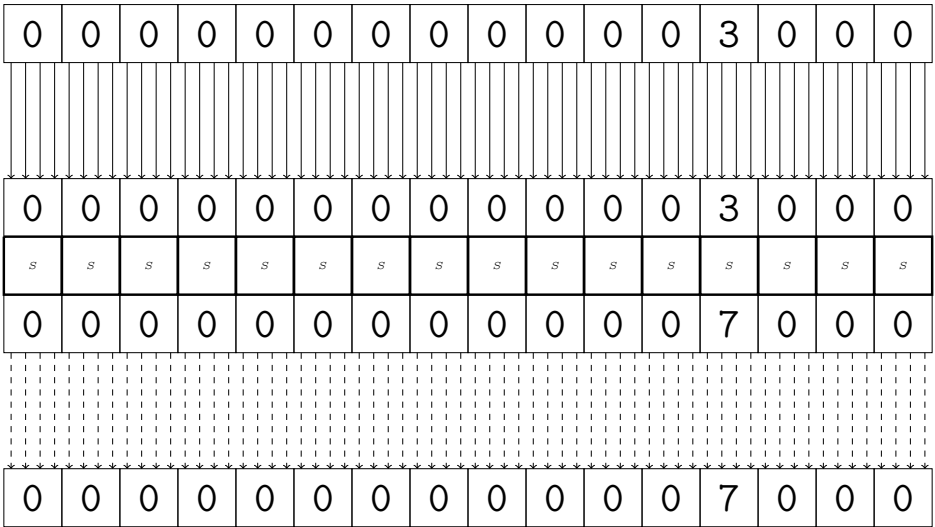
4 warp_round

Maximal linear correlation: $2^{-1.0}$

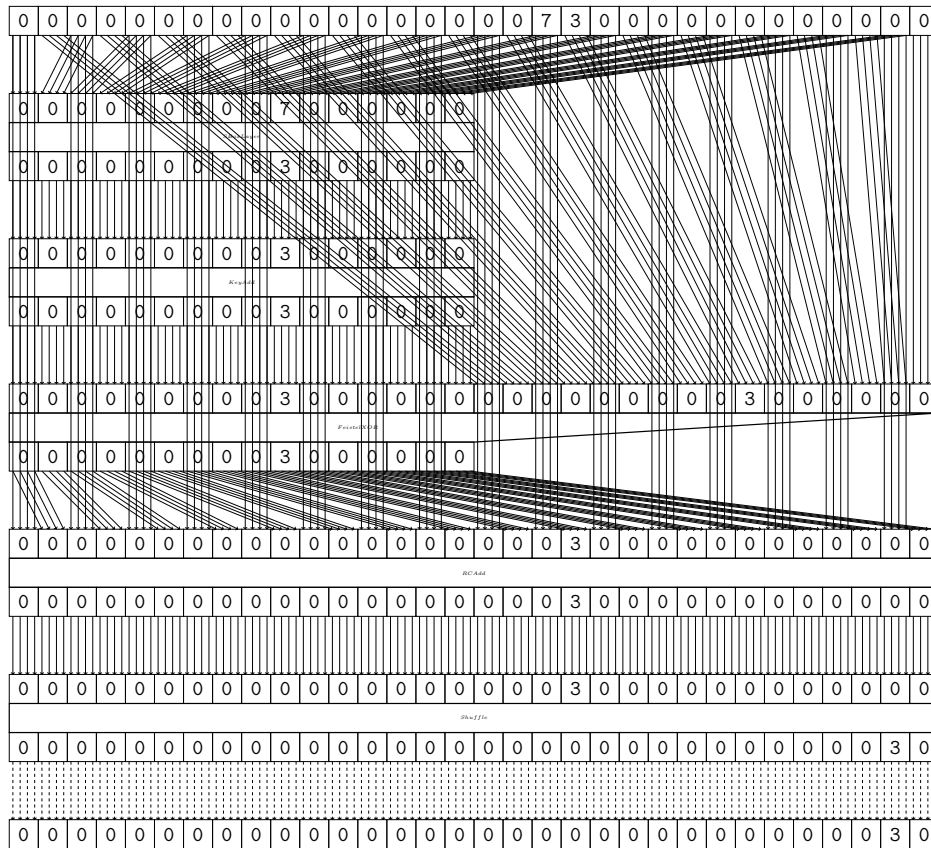


5 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

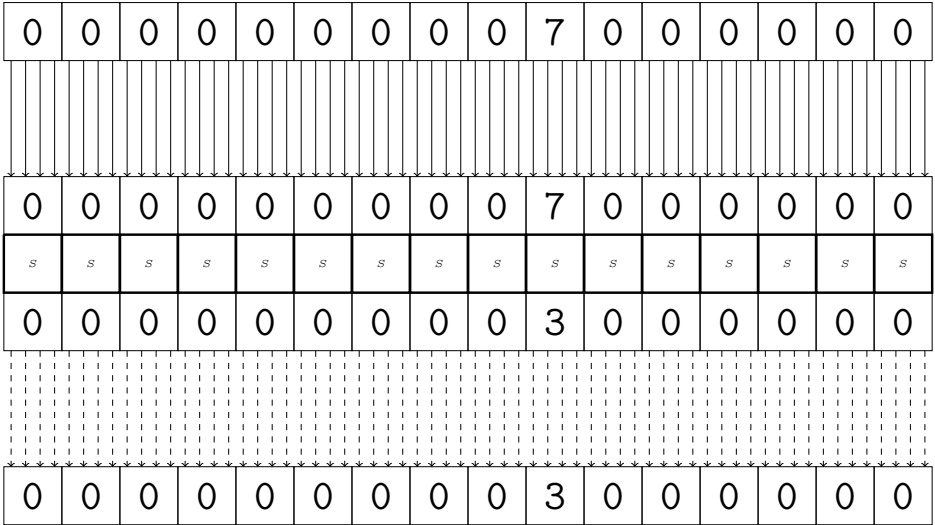


6 warp_round

Maximal linear correlation: $2^{-1.0}$ 

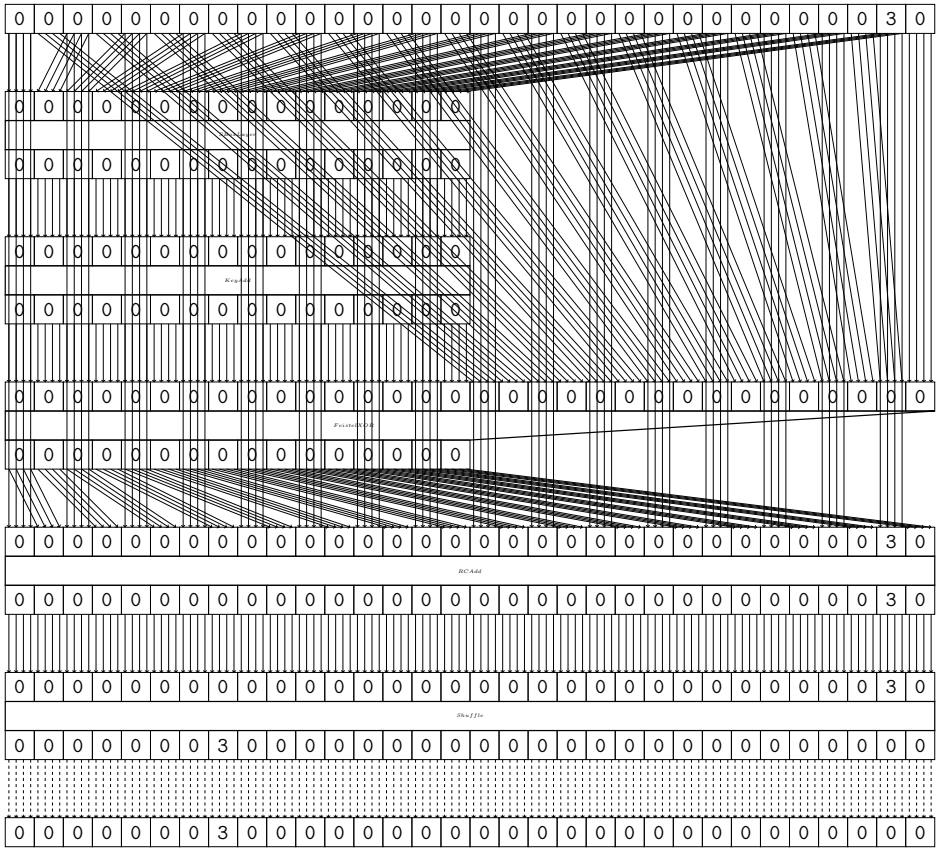
7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



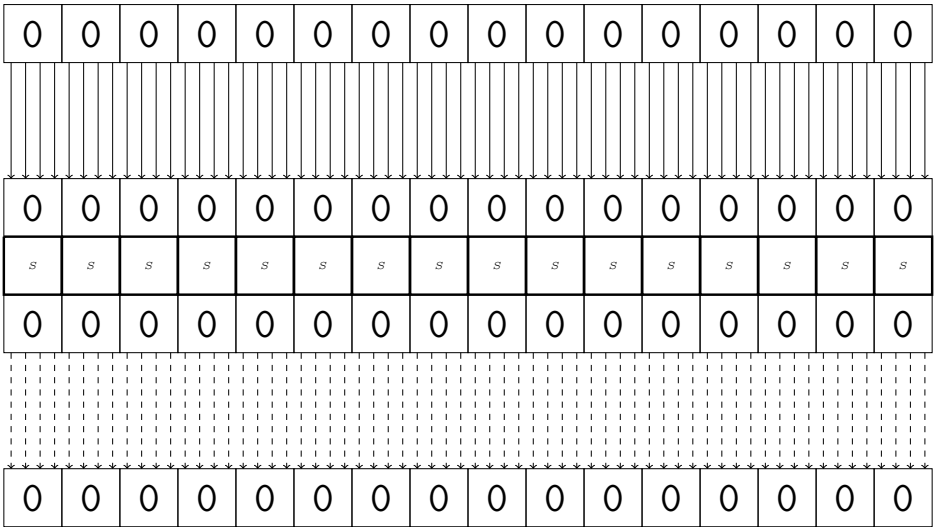
8 warp_round

Maximal linear correlation: $2^{-0.0}$

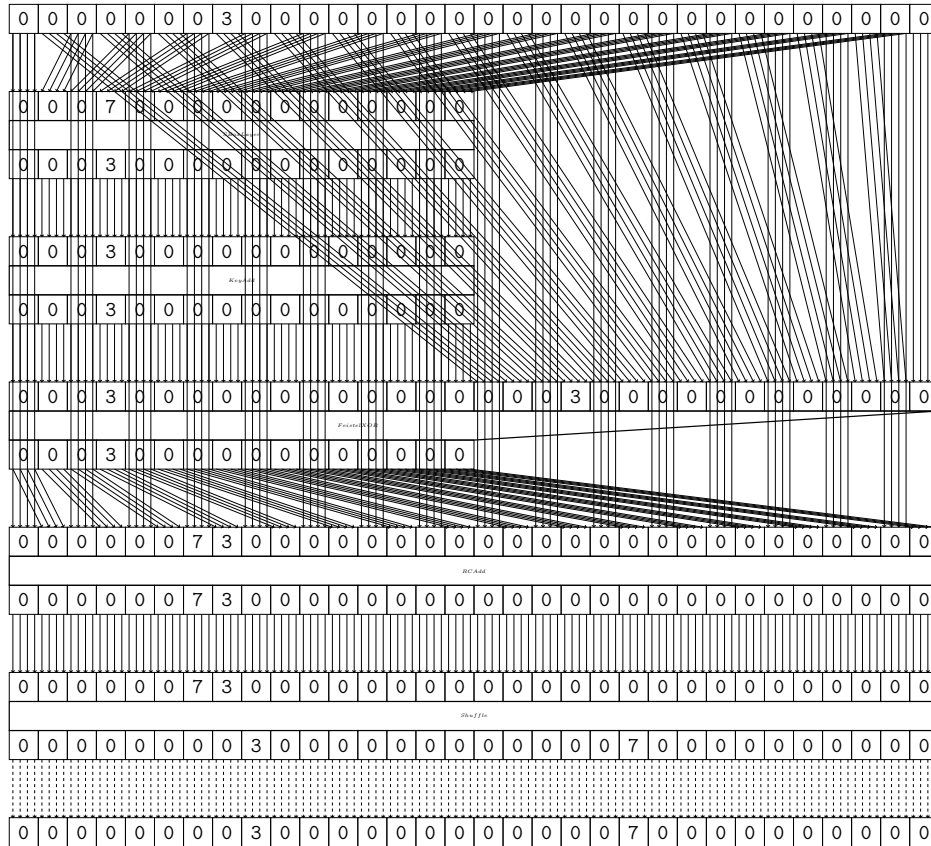


9 SBoxLayer

Maximal linear correlation: $2^{-0.0}$

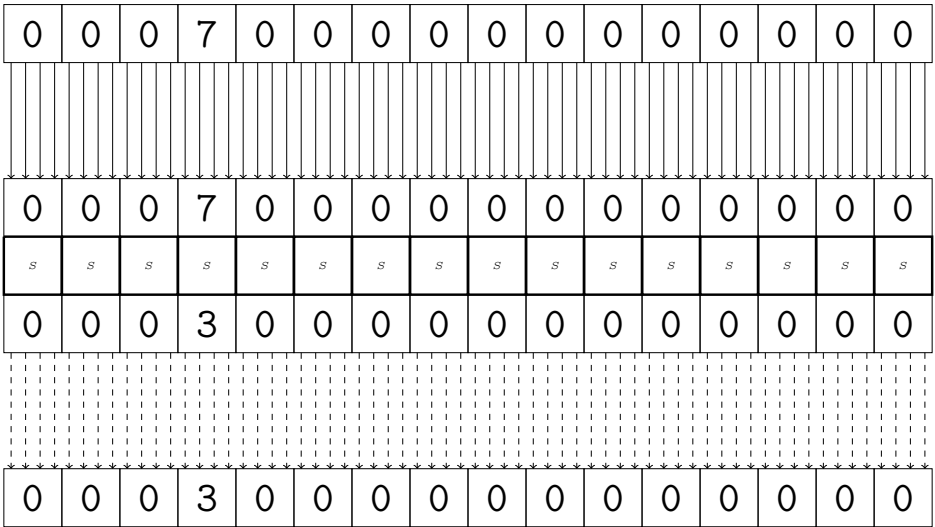


10 warp_round

Maximal linear correlation: $2^{-1.0}$ 

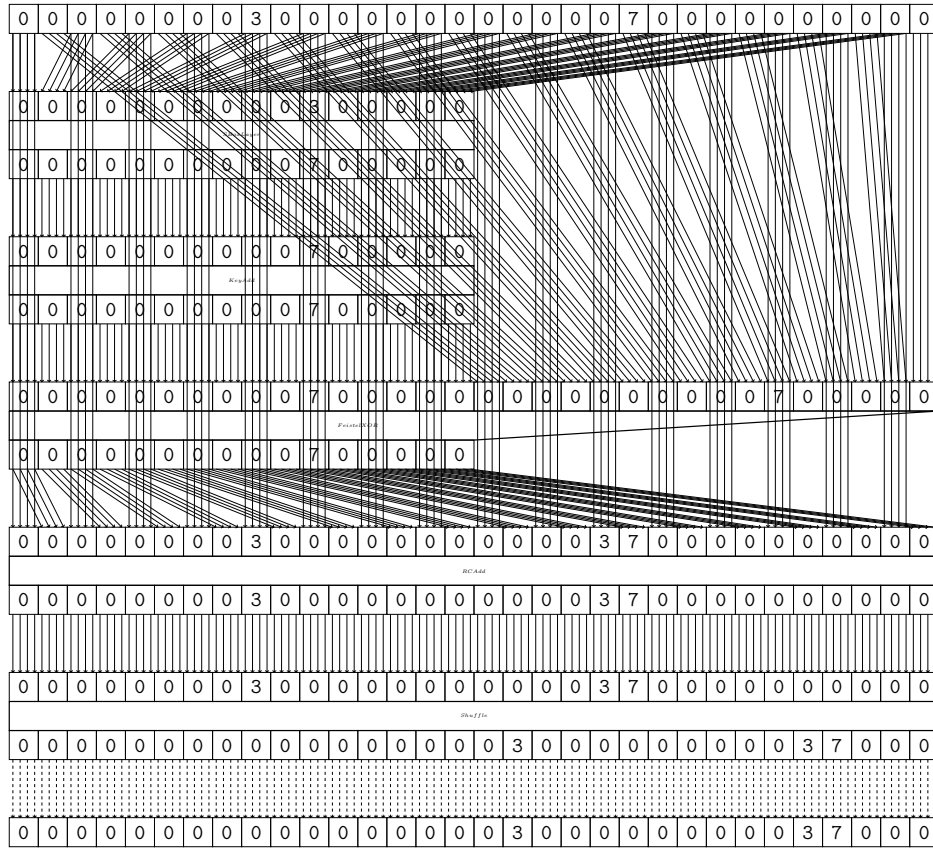
11 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



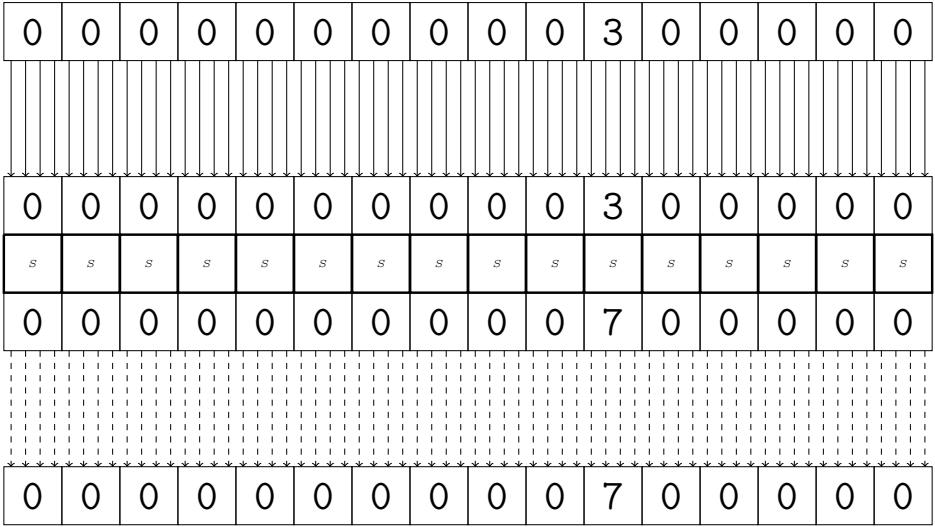
12 warp_round

Maximal linear correlation: $2^{-1.0}$



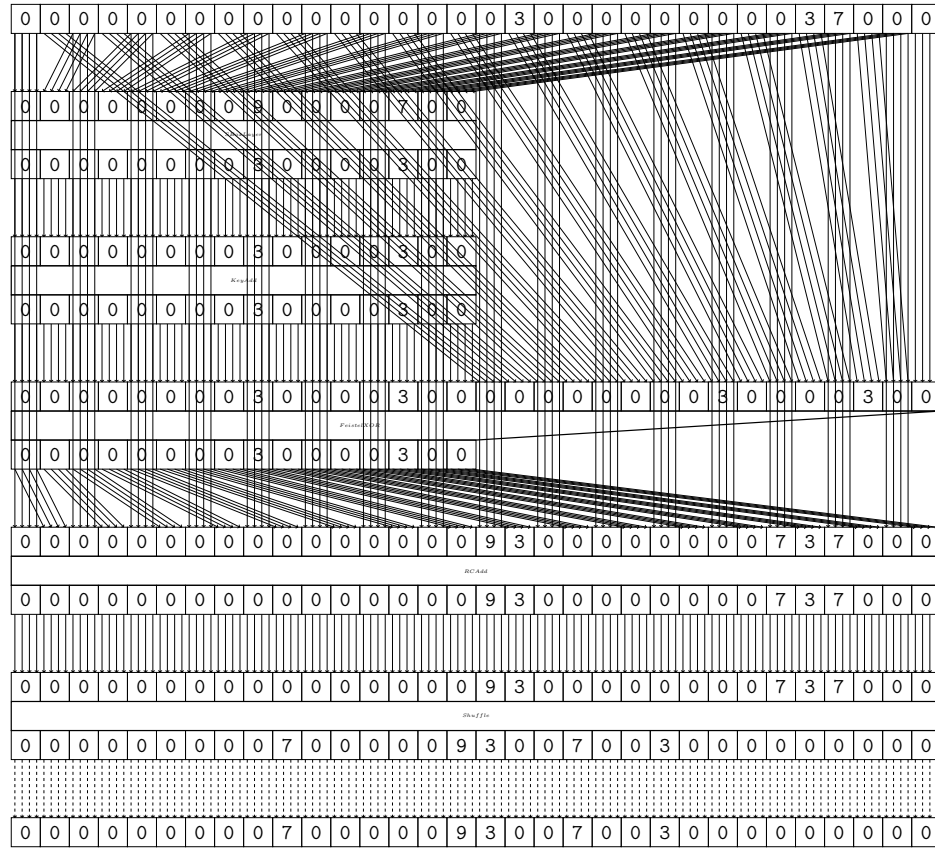
13 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



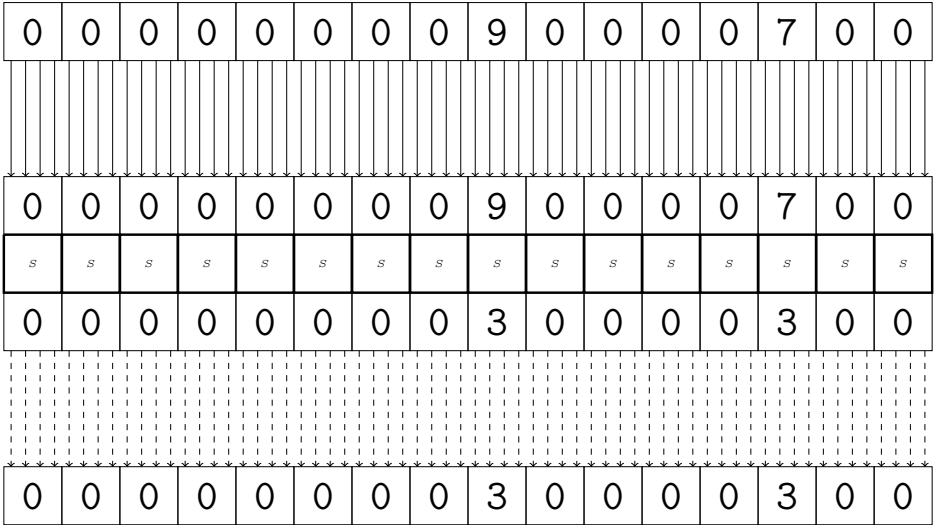
14 warp_round

Maximal linear correlation: $2^{-2.0}$



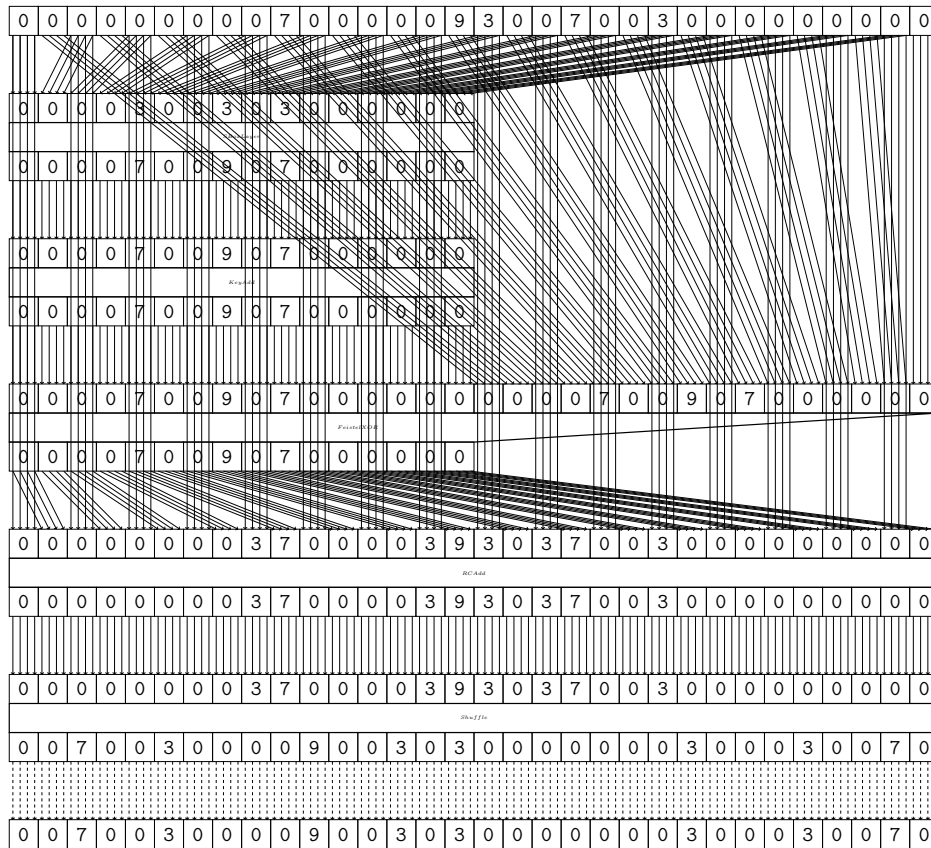
15 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



16 warp_round

Maximal linear correlation: $2^{-3.0}$



17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

